ECE1512 Project A

Whole-Slide Image Multiple Instance Learning

An ABMIL training pipeline in PyTorch

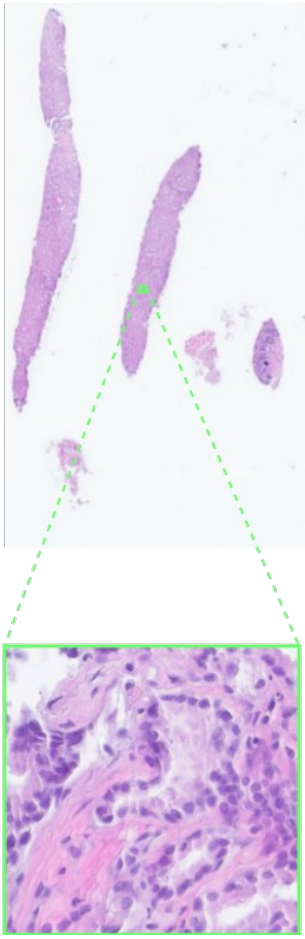
Linfeng Ye, October 7, 2026

Learning Objectives

- Understand the fundamental concepts of Whole-Slide Images.
- Learn how to build customized models in PyTorch and implement a complete supervised training pipeline.
- Develop the ability to formulate meaningful research questions and design practical approaches to investigate and solve them.

Whole-Slide Images (WSIs)

- WSIs are ultra-high-resolution digital scans of a pathology slide. WSIs support disease diagnosis and treatment planning, while also enabling remote consultation and collaboration between pathologists.
- Detecting a small cluster of cancerous cells in a gigapixel WSI can be like **finding a needle in a haystack**. This labor-intensive task places a heavy burden on a healthcare workforce already facing a shortage of pathologists [1].
- As computer vision engineers, **how can we use AI to support pathologists' diagnostic decisions and reduce their review workload?**



[1] Bychkov, A., & Schubert, M. (2023). Constant Demand, Patchy Supply. The Pathologist, February 17, 2023.

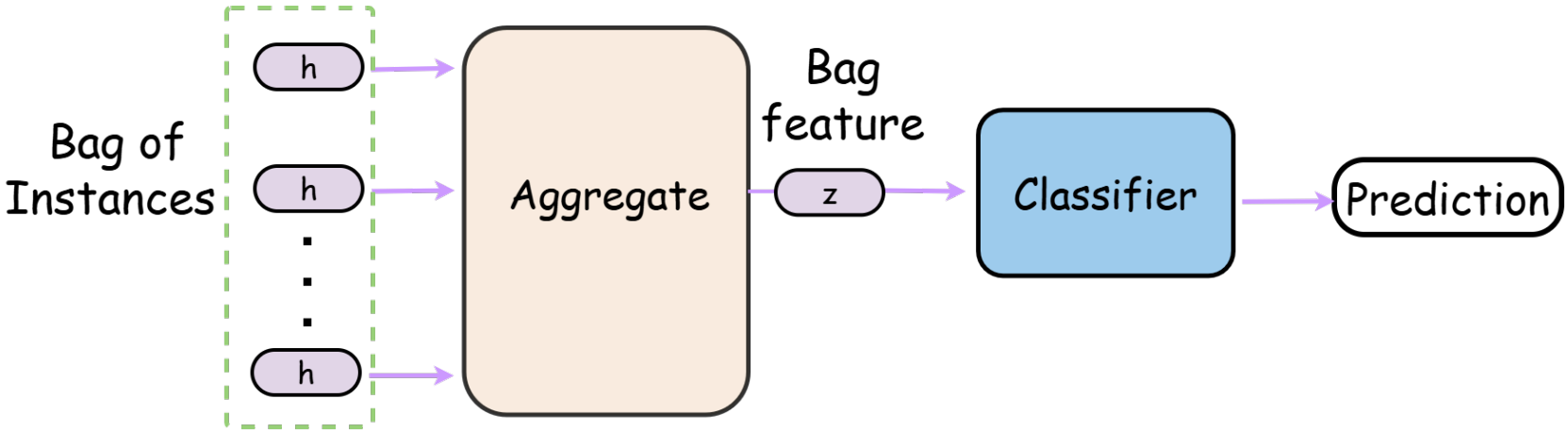
Multiple Instance Learning (MIL)

Setting

Bag of instances: Each sample is a bag $H = \{h_1, h_2, \ldots, h_n\}$, containing instances.

Bag-level supervision: Each bag has one ground-truth label (y), while individual instance labels are unknown.

Bag labeling assumption: A bag is labeled positive if at least one instance is positive, and negative if all instances are negative.



Why WSI Classification Fits MIL

WSI classification

To process WSIs, we typically divide them into **patches.**

A **diagnosis** per whole slide.

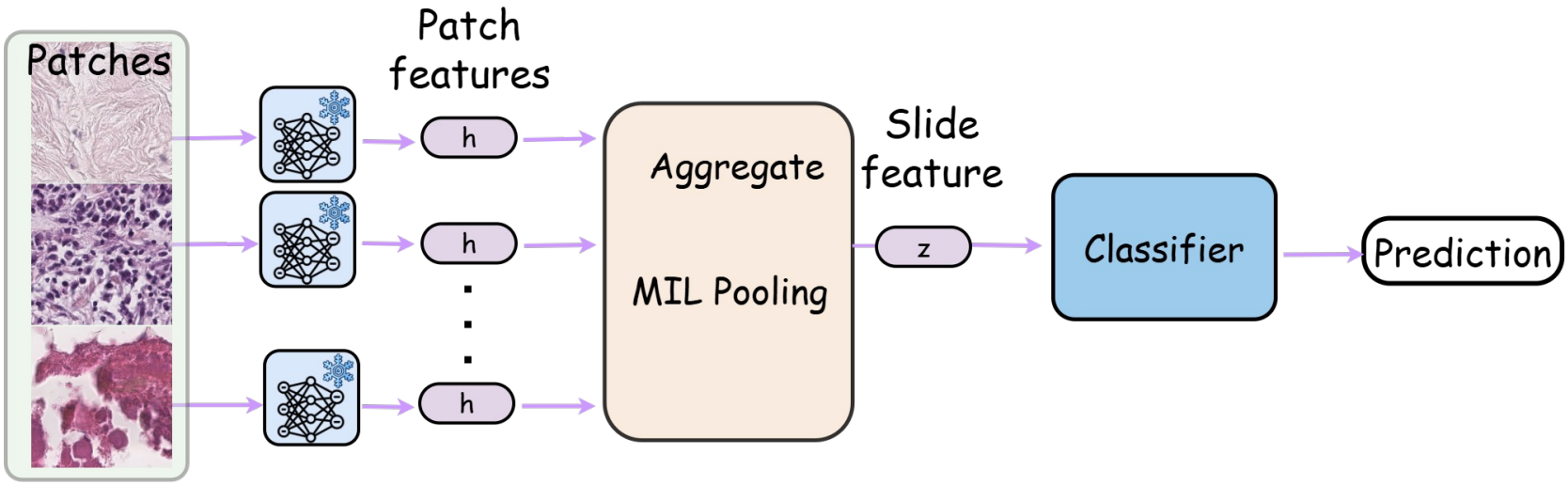
Slides contain **different numbers of patches**. A cancerous slide can also contain benign tissue.

MIL formulation

A **bag** contains multiple instances.

A bag has **one observed label**.

Bags can **vary in size** and contain mixed instances.



Fixed MIL Aggregation

Mean and max pooling are task and input agnostic.

Let $h_i \in \mathbb{R}^d$ be the feature vector of patch i, and N the number of patches.

Mean pooling

$$\mathbf{z} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{h}_i$$

Gives each instance equal weight. Small cancerous regions may contribute little.

Max pooling

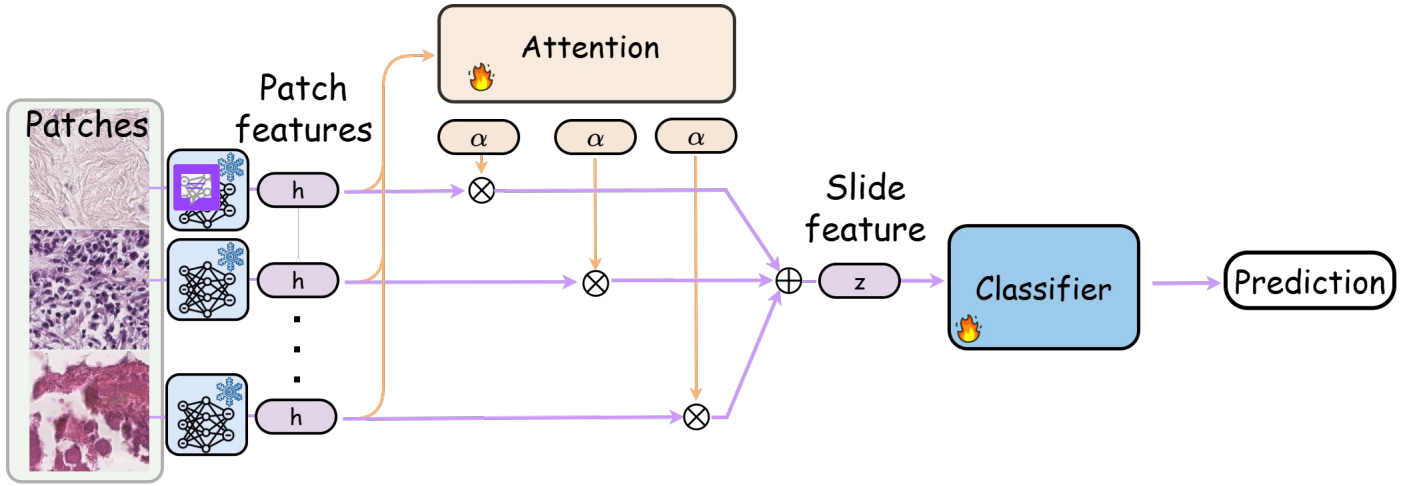
$$z_j = \max_{i=1,\dots,N} h_{ij}\,, \qquad j = 1, \dots, d$$

Selects the largest value in each feature dimension. Different dimensions may select values from different patches.

Attention-Based MIL

Fixed pooling cannot learn task-specific instance importance. ABMIL uses gated attention to score each patch and generate bag-dependent aggregation weights:

$$\mathbf{z} = \sum_{i=1}^{N} \alpha_i \, \mathbf{h}_i, \qquad \alpha_i = \frac{e^{s_i}}{\sum_{j=1}^{N} e^{s_j}}, \qquad s_i = \mathbf{w}^\top [\tanh(\mathbf{V}\mathbf{h}_i) \odot \sigma(\mathbf{U}\mathbf{h}_i)],$$



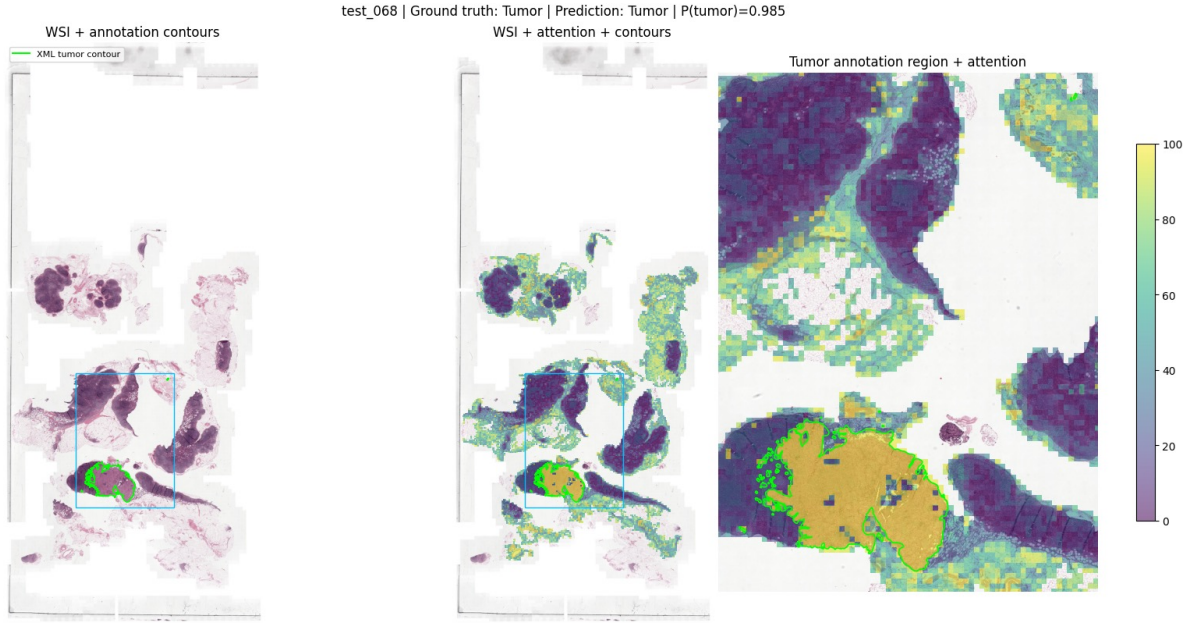
The attention module and classifier learn jointly from slide-level labels.

Attention Maps for Tumor Localization

Beyond classification, attention provides a way to visualize patch importance.

Map each patch's attention weight to its location in the WSI to highlight suspected cancerous regions.

Regions with high attention can highlight candidate cancerous areas for further review.



Representative MIL Methods

ABMIL has inspired extensive research on MIL for WSI analysis. Below are several representative methods.

- **CLAM** (Nature BME 2021)**:** Addresses limited labeled slides by combining attention pooling with instance-level clustering to improve feature discrimination.
- **DSMIL** (CVPR 2021)**:** Addresses sparse positive patches through dual-stream aggregation, combining the highest-scoring patch with other patches according to their similarity.
- **TransMIL** (NeurIPS 2021)**:** Addresses overlooked patch relationships through Transformer self-attention and positional encoding, incorporating morphological correlations and spatial context.
- **DTFD-MIL** (CVPR 2022)**:** Addresses having few labeled slides but many patches by creating pseudo-bags and distilling their features for a second MIL stage.

You are encouraged to explore additional MIL literature.

Assignment Outline 1

Literature review and research questions

- Read ABMIL, CLAM, DSMIL, TransMIL and DTFD-MIL.
- For each paper, summarize its contributions and the problem it addresses.
- Compare the methods and identify meaningful research questions.
- **Use AI to accelerate reading and explain unfamiliar concepts. Verify your summaries against the original papers.**

How to Read a Paper, S. Keshav, http://ccr.sigcomm.org/online/files/p83-keshavA.pdf

Assignment Outline 2

Complete the provided Jupyter notebook

- Complete every TODO cell in the provided notebook.
  - Build a customized MIL model using PyTorch.
  - Implement supervised training, validation.
  - Train the MIL model and report its test performance.
  - Visualize candidate cancerous regions using attention.

Expected outcome: independently build, train, and visualize a MIL model in PyTorch.

Assignment Outline 3

Propose two research ideas and implement one

- State the research question for each idea and justify your design choices.
- Possible directions: **explainability**, advanced model **architectures**, or better **objective functions** than cross-entropy.
- Compare the implemented method with ABMIL using the same data splits and metrics.
- Report your experimental observations and justify it.
- Negative results will not be penalized if your design choices and analysis are well justified.

Deliverables

Report, results, and code

Due Nov 4, 2026, via Q. Up to 20 pages: Paper summary + baseline results + two extension proposals + implementation and evaluation of one selected extension. Code MUST BE submitted for Project A.

Presentation materials

Due Dec 4, 2026, via Q. You will explain the approach in the solution and justify the results obtained.

Prepare a poster based on your project results and submit it via Quercus. The maximum dimensions are 5120 × 2880 pixels, and the file size must not exceed 10 MB.

Presentation: December 16, 2026